

David Marr

Systems Engineer — Backend, Infrastructure & AI Platform Primitives

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Summary

Backend and infrastructure engineer who builds the substrate underneath experiments — APIs, service boundaries, databases, queues, containers, and deployment pipelines that make prototypes runnable in production-like conditions. Twenty-five years on Python, Node.js, PostgreSQL, Docker, Kubernetes, and AWS. I wire models, agents, and services together, operate a home Kubernetes cluster to stay hands-on with networking and storage, and treat agentic AI as a systems primitive rather than a UI feature.

Experience

Context Labs, Principal Software Engineer (Platform & Infrastructure) – Cambridge, MA June 2022 – Mar 2026
Owned backend platform infrastructure for a multi-tenant analytics product — service integration, data access, auth, and operational reliability across distributed systems.

- Built production integration layer in Node.js/TypeScript (GraphQL BFF) wiring Java APIs, PostgreSQL, auth, tenancy, and localization into a stable backend surface other teams could build against.
- Defined service boundaries, API contracts, and deployment patterns for a distributed platform; improved observability and cut incident recovery time by 50%.
- Established platform primitives for data access and service orchestration so experiment teams spent less time re-solving plumbing.
- Drove adoption of AI coding assistants and MCP server tooling as infrastructure — agent interfaces, review guardrails, and repeatable patterns for safe experimentation on real systems.
- Operated multi-tenant backend services where correctness, access control, and uptime mattered as much as feature velocity.

Skills used: Python, Node.js, TypeScript, GraphQL, PostgreSQL, Trino, Docker, Kubernetes, Bash

Filtered.ai, Lead Engineer – Boston, MA Feb 2020 – Feb 2022
Lead backend engineer wiring models, queues, and data pipelines into a hiring-assessment platform on AWS and PostgreSQL.

- Built ingestion, scheduling, grading, and workflow services on AWS and PostgreSQL — the backend machinery connecting product logic to running infrastructure.
- Integrated computer-vision models and human-in-the-loop validation (AWS Mechanical Turk) into batch and on-demand scoring pipelines.
- Owned model deployment hooks, data validation, and operational monitoring for automated assessment at scale.

Skills used: Python, AWS, PostgreSQL, Node.js, Bash

Turo, Staff Software Engineer – Cambridge, MA Apr 2018 – Dec 2019
Owned backend systems for trip management and dynamic pricing on PostgreSQL — the machinery behind product-facing availability and rate logic.

- Owned backend services for dynamic pricing and trip management on PostgreSQL — availability rules, rate logic, and transactional consistency under load.
- Maintained host-recognition and incentive systems backed by dependable data pipelines and service-level operational behavior.

Skills used: PostgreSQL, TypeScript, JavaScript

OneMarket / Westfield Labs, Senior Software Architect – San Francisco, CA Jan 2016 – May 2018
Integrated hardware, payments, and backend services into a smart parking platform — cameras, billing, gate controllers, and venue APIs coordinated through PostgreSQL and Node.js services.

- Architected service and data flows across cameras, payment rails, gate hardware, and billing systems — ~\$2.7m in monthly recurring revenue depended on reliable backend orchestration.
- Built low-level device integration in C alongside API and database services supporting 60,000+ monthly sessions across 4 flagship venues.

Skills used: Node.js, PostgreSQL, C, Stripe, Bash

Tilt.com, Senior Systems Engineer – San Francisco, CA July 2013 – Dec 2015

- Built and operated Perl/Dancer backend services with PostgreSQL — request handling, data modeling, and server-side rendering integration for a high-traffic payments product.
- Iterated contribution-flow backend logic yielding 15% lift in site signups through simpler service paths and data handling.

Skills used: Perl, PostgreSQL, JavaScript, Bash

Various, Previous Leadership & Technical Roles – Massachusetts & California 2000-2013

- Enterprise web integrations and industrial automation backends (GE Fanuc / Intellution) — device data, dashboards, and service plumbing.
- Publishing and ad-tech platform services at Say Media; co-founded early social product (Secret Feature).
- High-traffic backend and integration work for Yahoo properties; design-to-production systems prototyping at Frog Design.

Skills used: Node.js, MySQL, PHP, .NET, SQL Server, Bash

Systems

Personal homelab (Kubernetes / GitOps)

Self-hosted infrastructure lab documented on marr.github.io — Talos Linux, Flux, SOPS secrets, and service orchestration on a home cluster.

- Operate Talos Kubernetes with Flux GitOps, SOPS secrets, and persistent storage classes — hands-on with containers, networking, reconciliation, and failure recovery.
- Built observability, service health checks, and agent-oriented runbooks for a multi-service home environment documented publicly.

Education

University of New Hampshire, BS in Business Administration – Durham, NH Sept 1996 – May 2000
Focus on Management of Information Systems (MIS)

Skills

Infrastructure & Backend: Docker, Kubernetes, AWS, Python, Node.js, PostgreSQL, GraphQL, Trino, Bash, Git, MongoDB, Go

AI, Agents & Services: MCP, Vercel AI SDK, Cloudflare Workers, Cloudflare Durable Objects, Agent Skills, Cursor, Opencode, Pi

Systems Integration: TypeScript, C, Perl, Stripe, WebRTC, networking, service orchestration, observability